

Multilevel Logistic Regression on NBA Shot Data

Ji Wang
Hanchen Qiu

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1 Introduction

NBA shot outcome modeling is used to understand and predict whether a field goal attempt is made or missed, especially at different distances. This study investigates the relationship between goal rate and several key factors, and discusses whether the effect of distance varies significantly across different players. By incorporating individual factors into statistical modeling, such study may provide quantitative insights of players' capability, which might enable coaches and teams to make more reasonable decisions. Due to the binary nature of shot data, logistic regression is used for modeling, with control variables included.

Individual difference contributes to the hierarchical data structure. Although standard logistic regression model ignores individual differences, Bayesian multilevel logistic regression model is suitable to handle it. Within such models, player-specific slopes are used for distance to respond to individual differences, while fixed slopes are introduced for control variables. However, it is assumed that individuals' reaction to all factors are constant across the 2014-2015 NBA season.

This approach is also useful for low-volume shooters, whose raw percentages can be unstable. Through partial pooling, the model shrinks noisy player estimates toward the league average while still preserving meaningful individual differences. The Bayesian framework also provides full posterior distributions, allowing uncertainty to be quantified and players to be compared probabilistically after adjusting for shot context.

The model assumes that each player's shooting ability and response to shot-context variables are constant across the 2014-15 NBA season. This is a limitation because player performance can vary over time due to shooting form, fatigue, injuries, changing team roles, or coaching strategy. Therefore, the model captures average season-level player effects rather than time-varying performance.

Furthermore, it also omits some important basketball context, such as court location and live score margin. In particular, the clutch variable is only an approximation because the dataset contains final game margin instead of actual score margin at the time of the shot.

Section 2 of this report will describe the dataset in detail and define research questions. Section 3 will introduce the model design and 2 priors used. Results will be presented in section 4 with analysis about impact of different priors.

2 Data Description

The dataset is Kaggle NBA shot logs, which records every regular-season shot attempt in the 2014-2015 NBA season.

2.1 Variables' Definition

Variable	Description
Shot_result / FGM	Binary indicator of whether the field goal attempt was made. A value of 1 represents a made shot, and 0 represents a missed shot.
Shot_dist	Distance from the basket to the shot location, measured in feet.
Close_def_dist	Distance from the shooter to the closest defender, measured in feet.
Shot_clock	Number of seconds remaining on the shot clock when the shot was attempted.
Period	The game period in which the shot occurred, including regulation quarters and overtime periods.
Game_clock	Time remaining in the period, converted from MM:SS format to seconds.
Final_margin	Final score margin of the game.
Clutch	Binary indicator for shots taken in the fourth quarter or overtime with five minutes or less remaining in games with an absolute final margin of five points or fewer.

Table 1: Description of Key Variables

Variables mentioned above play important roles in our research. Other variables in the original dataset, such as matchup and game result, were retained for reference but not included in the final model.

2.2 Data Cleaning & Feature Engineering

Data cleaning is done considering the basketball context. Observations with missing shot block values are deleted. Implausible values, such as negative

distance measurements and negative touch time. To pursue reliable estimation regarding individual effects, only players with at least 100 shot attempts were remained in this study.

Features are also engineered thanks to the basketball context. The clutch indicator is designed to represent intense late-game scenarios. A shot was recorded as clutch if it satisfies three conditions simultaneously:

- It occurred overtime or in the fourth quarter;
- It occurred with five minutes or less before the session ended;
- The absolute final score margin is fewer or equal to five

Finally, continuous predictors were standardized before model fitting, so that coefficients could be compared on the same dimension. The cleaned dataset was then split into training and testing according to the 80/20 rule. Player-level stratification was used in data splitting.

2.3 Descriptive Statistics

There are 121,617 shot attempts from 274 players in the cleaned dataset. The overall success rate is 45.6%, with clutch shots account for 3.4% of all the shots. Table 1 summarizes main descriptive statistics for cleaned dataset. Additional exploratory figures, such as shot distance distribution, are provided in Appendix A.

Statistic	Value
Number of shots	121,617
Number of players	274
Overall field goal percentage	45.6%
Number of clutch shots	4,094
Share of clutch shots	3.4%
Mean shot distance	13.46 ft
Median shot distance	13.50 ft
SD of shot distance	8.77 ft
Mean closest defender distance	4.12 ft
Median closest defender distance	3.70 ft
SD of closest defender distance	2.75 ft
Mean shot clock	12.45 sec
Median shot clock	12.30 sec
SD of shot clock	5.75 sec
Mean touch time	2.77 sec
Median touch time	1.60 sec
SD of touch time	2.94 sec

Table 2: Descriptive statistics of cleaned NBA shot log dataset

2.4 Research Questions

This research focuses on five research hypotheses:

- H1: Shot distance is negatively associated with shot result;
- H2: Closest defender distance is positively associated with shot result;
- H3: Clutch-time shots have lower field goal percentage than regular shots;
- H4: Player-level random intercept variance is significantly > 0 ;
- H5: The effect of distance varies significantly across players.

3 Model Design and Prior Specification

3.1 Model Design

Since the response variable is binary, the model is designed as a Bayesian logistic regression. Let Y_{ij} denote the outcome of shot i taken by player j , with $Y_{ij} = 1$ if the shot is made and $Y_{ij} = 0$ otherwise. Then the response variable is treated as a Bernoulli random variable:

$$Y_{ij} \sim \text{Bernoulli}(p_{ij})$$

where p_{ij} represents the probability that the shot is made.

The regression formula is:

$$\begin{aligned} \text{logit}(p_{ij}) = & \beta_{0j} + \beta_{1j}\text{ShotDist}_{ij} \\ & + \beta_2\text{DefDist}_{ij} + \beta_3\text{ShotClock}_{ij} + \beta_4\text{Clutch}_{ij}. \end{aligned}$$

Shot distance, closest defender distance, and shot clock time are standardized continuous variables. Clutch is a binary variable. β_{0j} is the player-specific intercept used to capture player j 's baseline shooting capability. Coefficient β_{1j} is the player-specific slope for shot distance, which may differ across different players. The remaining coefficients are fixed effects shared by everyone. They are designed to measure the average association between shot success and shot clock time, closest defender distance, and clutch scenarios.

Player-specific intercepts and distance slopes are assumed to come from a common population distribution, which leads to the hierarchical model structure:

$$(\beta_{0j}, \beta_{1j}) \sim N((\gamma_{00}, \gamma_{10}), \Sigma).$$

Thanks to this population distribution, the population-average baseline shooting tendency is measured by γ_{00} , and the population-average effect of distance is represented by γ_{10} . Σ describes the variation across players in baseline shooting ability and distance sensitivity.

Distance slope is allowed to vary individually due to the basketball context. For instance, interior players may experience sharper decline in shooting probability when distance increases, while perimeter shooters might be less affected. To ensure that the model is stable and interpretable, other contextual variables are treated as fixed effects across players.

Partial pooling is also enabled by the hierarchical structure. Player-specific estimates are shrunk toward the population average. This effect becomes even more significant when a player has fewer shot attempts recorded. This helps to provide more reliable estimates.

3.2 Prior Specification

3.2.1 Prior A: weakly informative prior

Prior A is designed to be weakly informative, which regularizes the parameters without strongly shrinking posterior estimates in a particular direction. Thus, the data can play the main role when determining regression coefficients, while still avoiding extreme or unrealistic parameter values.

The model uses:

$$\gamma_{00} \sim N(-0.2, 1)$$

The league-average goal rate is about 45%, corresponding to $\text{logit}(0.45) \approx -0.2$.

For the population distance slope and fixed-effect coefficients, the prior mean is set to be 0:

$$\gamma_{10}, \beta_2, \beta_3, \beta_4 \sim N(0, 0.5).$$

Such priors reflect the weakly informative purpose. Centering the priors at 0 suggests that the model does not assume any effect before observing the data. The standard deviation of 0.5 allows meaningful effects on the final result, but discourages implausibly large coefficients. [1]

For player-level random effects, their standard deviations are assigned a half-normal prior distribution:

$$\sigma_{\beta_0}, \sigma_{\beta_1} \sim \text{HalfNormal}(1).$$

Such priors ensure that the standard deviations are positive and enable the model to measure individual differences [2].

The correlation between the player-specific intercept and distance slope is modeled using an LKJ prior with $\eta = 2$. This prior is a standard solution for correlation matrices in Bayesian hierarchical models. [3] Setting $\eta = 2$ mildly regularizes the correlation matrix toward the identity matrix, discouraging extreme correlation if not strongly supported by the data. Since there is no strong prior evidence that baseline shooting capability and distance sensitivity are highly correlated, this setting is appropriate.

In short, Prior A provides a flexible but stable baseline model. While maintaining reasonable regularization for Bayesian computation, it gives the data freedom to estimate the main shot-quality effects and individual-level variation.

3.2.2 Prior B: strongly informative based on literature

Prior B integrates previous quantitative findings for basketball analytics directly into the Bayesian model, providing a highly informative prior. This prior anchors the parameters to be close to the values calculated in the literature.

The model uses:

$$\gamma_{00} \sim N(-0.18, 0.2)$$

The intercept is tightly centered on $\text{logit}(0.455) \approx -0.18$, matching the exact observed league-average field goal rate of 45.5%. The narrow standard deviation of 0.2 reflects high confidence that the population-level baseline lies within a small neighborhood of the league mean.

In terms of the population-level distance slope and fixed coefficients, the prior means are set based on published studies:

$$\gamma_{10} \sim N(-0.07, 0.02), \quad \beta_2 \sim N(0.05, 0.02),$$

$$\beta_3 \sim N(0.02, 0.01), \quad \beta_4 \sim N(-0.10, 0.05).$$

The mean of distance slope is set around -0.07, encoding the well-studied negative effect of shot distance on FG probability [4] [5], while the positive prior of +0.5 reflects the intuition that greater defender separation improves shot success. The small positive prior on shot clock (+0.02) captures the modest benefit of having more time to set up the shot, while the negative prior for the clutch indicator (-0.10) demonstrates that the stress of late-game leads to worse shooting performance. [6] The deviation of clutch coefficient is set to be wider than that of distance and defender effects, which have already been well established.

For player-level random effects, their standard deviations are assigned tighter half-normal priors:

$$\sigma_{\beta_0}, \sigma_{\beta_1} \sim \text{HalfNormal}(0.4).$$

Such priors ensure that the standard deviations remain positive while reflecting the empirical pattern that most NBA players' shooting ability cluster within a narrow range.

The correlation between the player-specific intercept and distance slope is modeled using an LKJ prior with $\eta = 4$, which provides a stronger regularization for the correlation matrix towards the identity matrix compared with Prior A's $\eta = 2$. This choice reflects a stronger belief that a player's baseline shooting ability is largely independent of their sensitivity to distance.

In short, Prior B provides a domain-knowledge-driven model in which prior research plays a more important role in the posterior distribution. Although still providing flexibility, the fixed effects are tightly anchored near the calculated values, serving as a comparison with Prior A.

4 Analysis results

This part will introduce the detail in model training and diagnostics first. After that, fixed effects and player-specific random effects will be covered. Then prior sensitivity will be analyzed to prepare for model comparison. Furthermore, this section will validate predictions before summarizing all the findings.

4.1 MCMC Sampling Setup and Chain Tuning

Two versions of the model were fitted using weakly informative prior A and stronger informative prior B. For each version, the model was sampled by four Markov chains. Each chain contains 1,000 tuning iterations with 2,000 posterior draws, leading to 8,000 posterior draws in total across the four chains.

No-U-Turn sampler (NUTS) is used for model fitting, which is the default Hamiltonian MC sampler in PyMC package. Target acceptance rate is 0.95 for avoiding divergent transitions, which can occur in hierarchical Bayesian models with correlated random effects.

The player-level random effects were implemented using a non-centered parameterization. Instead of sampling individuals' intercept and distance slope directly from multivariate normal distribution,

$$\begin{pmatrix} \beta_{0j} \\ \beta_{1j} \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_{\beta_0} \\ \mu_{\beta_1} \end{pmatrix}, \Sigma \right),$$

the model first sample standardized player offsets:

$$z_j \sim N(0, I),$$

then transforms them using Cholesky factor of random-effect covariance matrix:

$$\Sigma = LL^T,$$

where L is the Cholesky factor:

$$L = \text{chol}(\Sigma).$$

So the player-specific parameters are constructed as:

$$\begin{pmatrix} \beta_{0j} \\ \beta_{1j} \end{pmatrix} = \begin{pmatrix} \mu_{\beta_0} \\ \mu_{\beta_1} \end{pmatrix} + Lz_j.$$

This reparameterization separates the standardized player-level offsets from scale of random effects. Such technique is widely used in hierarchical Bayesian modelling since it reduces posterior dependence between group-level parameters and variance parameters and avoids divergent transitions. This research involves a random intercept and a random distance slope for each players, making such design quite beneficial.

Convergence and sampling quality evaluation are introduced in the following section.

4.2 Convergence Diagnostics

This section covers convergence diagnostics to examine whether the MCMC chains generated reliable posterior samples. Main evaluation matrix includes R-hat, effective sample size (ESS) and trace plots. R-hat evaluates whether the four MCMC chains mixed to the sample posterior distribution. ESS measures the amount of independent information included in the posterior draws. Divergent transitions were also examined to check problems with posterior geometry.

Parameter	Mean	SD	94% HDI	ESS Bulk	R-hat
μ_{β_0}	-0.2141	0.0102	[-0.2332, -0.1950]	6498.62	1.0002
μ_{β_1}	-0.5665	0.0131	[-0.5913, -0.5423]	5848.86	1.0001
β_2	0.3140	0.0091	[0.2967, 0.3311]	12811.87	1.0008
β_3	0.0994	0.0069	[0.0865, 0.1124]	15663.94	1.0018
β_4	-0.1430	0.0365	[-0.2110, -0.0740]	19492.70	1.0000
RE SD: β_{0j}	0.1012	0.0111	[0.0791, 0.1208]	3374.08	1.0016
RE SD: β_{1j}	0.1382	0.0116	[0.1159, 0.1591]	3002.39	1.0017

Table 3: Convergence diagnostics and posterior summary for Prior A

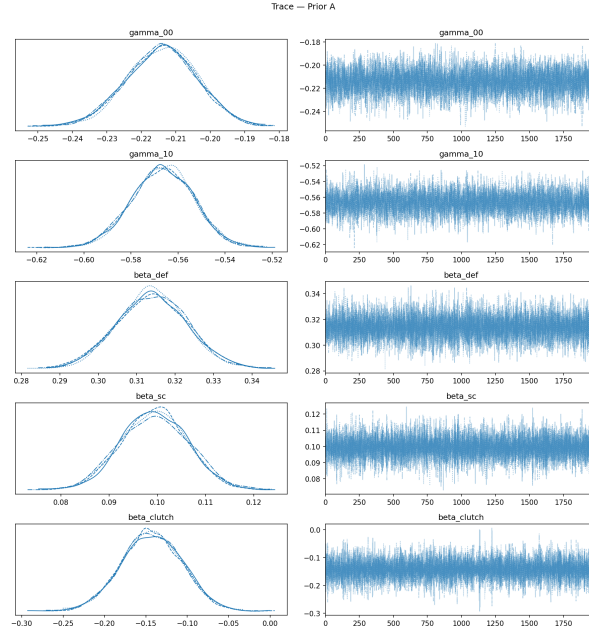


Figure 1: Trace Plot for Prior A

The trace plots Figure 1 also support convergence. The four chains overlapped closely for the main fixed-effect parameters and did not show significant

long-term trends or separation between chains. This suggests the chains explored same posterior regions rather than remaining stuck in different parts of the parameter space.

Convergence diagnostics of Prior B as also presented:

Parameter	Mean	SD	94% HDI	ESS Bulk	R-hat
μ_{β_0}	-0.2109	0.0117	[-0.2320, -0.1887]	796.74	1.0108
μ_{β_1}	-0.3697	0.0188	[-0.4032, -0.3324]	1165.32	1.0027
β_2	0.2475	0.0080	[0.2323, 0.2622]	7238.50	1.0009
β_3	0.0827	0.0056	[0.0722, 0.0932]	14478.56	1.0005
β_4	-0.1320	0.0296	[-0.1867, -0.0754]	14926.80	1.0004
RE SD: β_{0j}	0.0869	0.0111	[0.0661, 0.1076]	3352.19	1.0008
RE SD: β_{1j}	0.2062	0.0215	[0.1666, 0.2470]	831.76	1.0056

Table 4: Convergence diagnostics and posterior summary for Prior B

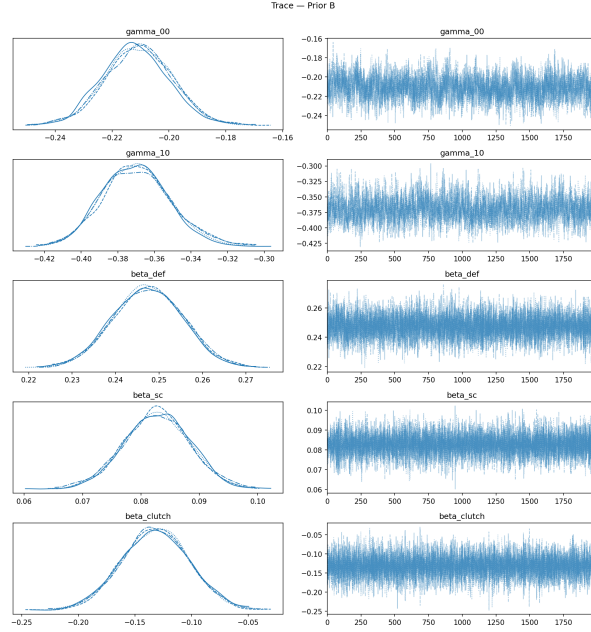


Figure 2: Trace Plot for Prior B

Still, most parameters under Prior B had R-hat values close to 1 and adequate ESS. However, Prior B showed slightly weaker convergence with R-hat of μ_{β_0} mildly above 1.01. Such breach may not affect the main interpretation of this study.

The diagnostics show that the primary model with Prior A converged quite well and generated reliable posterior samples. Therefore, Prior A is used for

main regression interpretation in the following sections, with B mainly used to evaluate prior sensitivity.

4.3 Posterior regression results

The primary model is based on Prior A with better convergence, so the main posterior regression results are reported using it. All regression coefficients will be interpreted on the log-odds scale due to the logistic link design. Continuous predictors were standardized before model fitting, so their coefficients represents the effect of one-standard-deviation increase. In the corresponding predictor on the log-odds of making a shot.

Parameter	Mean	94% HDI	Odds Ratio
μ_{β_0}	-0.2141	[-0.2332, -0.1950]	0.8072
μ_{β_1}	-0.5665	[-0.5913, -0.5423]	0.5675
β_2	0.3140	[0.2967, 0.3311]	1.3689
β_3	0.0994	[0.0865, 0.1124]	1.1045
β_4	-0.1430	[-0.2110, -0.0740]	0.8668

Table 5: Posterior regression results under Prior A

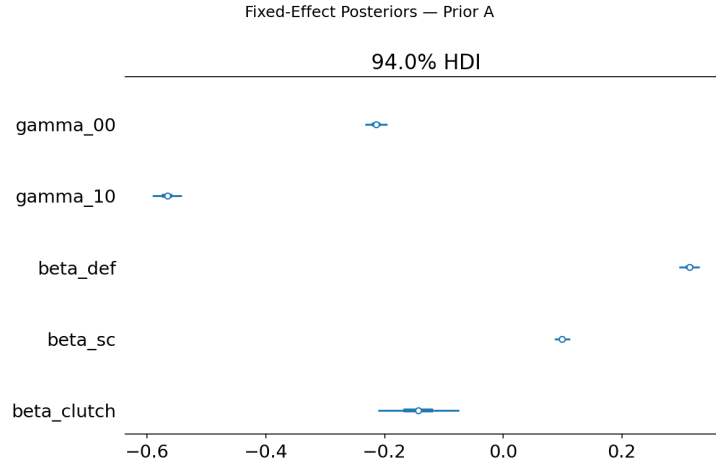


Figure 3: Trace Plot for Prior A

The posterior mean for the population intercept is -0.2141, corresponding to a baseline make probability of approximately 44.7% for an average player under average shot conditions and non-clutch status. This is close to the observed league field goal percentage in the cleaned dataset.

Posterior mean of shot distance slope is -0.5665 and its credible interval is entirely negative. This suggests that longer shots are substantially less likely to

be made. The odds ratio is approximately 0.57, indicating that one standard deviation increase in shot distance reduces the odds of making a shot by around 43%.

Defender distance has a positive effect, with a posterior mean of 0.314. The credible interval is entirely positive, showing that shots taken with more space from the nearest defender are more likely to succeed. The odds ratio is about 1.37, hinting that one standard deviation increase in defender distance increases the odds of making a shot by around 37%.

Shot clock has positive but narrower effect. The posterior mean is only 0.0994, leading to an odds ratio of 1.10.

The clutch coefficient is negative, with an entirely negative credible interval. The odds ratio is about 0.87, suggesting that clutch shots are associated with about a 13% decrease in the odds of making a shot when other factors controlled.

Overall, the posterior results support the main expectations of the model: longer distance reduces shot success, greater defender separation improves shot success, more shot-clock time has a modest positive effect, and clutch situations are associated with lower shooting success.

4.4 Player-level random effects

The multilevel model also estimates player-level random effects. Such random effects allow the model to interpret repeated shot attempts from each player to describe differences in player-specific tendencies.

Two kinds of random effects are included: player-specific intercept and shot distance slope, which has been carefully introduced in previous sections.

Random Effect	Mean SD	94% HDI	Interpretation
σ_{β_0}	0.1012	[0.0791, 0.1208]	Variation in baseline shooting tendency
σ_{β_1}	0.1382	[0.1159, 0.1591]	Variation in distance sensitivity

Table 6: Standard deviation estimation of two player-specific effects

The β_{0j} estimation of top 20 and bottom 20 players are presented in Appendix D.

The posterior mean of σ_{β_0} is 0.1012, with a credible interval entirely larger than 0. This shows that there is meaningful variation across players in baseline shooting tendency, even after controlling for shot context. In other words, different players do enjoy varying underlying shot-making ability.

The posterior mean of σ_{β_1} is 0.1382, also with a credible interval entirely larger than 0. This means that the effect of shot distance is heterogenous among players. Some players are more resilient when shot distance becomes longer. These results justify using a random distance slope in this model.

Last but not the least, the player-level random effects are estimated with partial pooling. As a result, player estimates are shrunk toward the population average, especially for players with fewer shot attempts. This let the player-level estimate become more stable than raw field goal percentages and relieve the risks of overfitting noisy individual performance.

4.5 Prior sensitivity analysis

This section conducts prior sensitivity analysis to examine whether the main conclusions depend on the choice of prior distributions. The primary model used weakly informative Prior A, and Prior B is a more informative prior based on expected directions according to basketball shot-quality research. This analysis compared posterior means, credible intervals, mean shifts and sign agreement between two priors for major parameters. Key results are summarized in Table 7:

Parameter	Prior A Mean	Prior B Mean	Direction Agreement
μ_{β_1}	-0.5665	-0.3697	Yes
β_2	0.3140	0.2475	Yes
β_3	0.0994	0.0827	Yes
β_4	-0.1430	-0.1320	Yes
σ_{β_0} SD	0.1012	0.0869	Yes
σ_{β_1} SD	0.1382	0.2062	Yes

Table 7: Parameter comparison based on different priors

These results show that the direction of main effects is consistent across two priors. Under both priors, shot distance has a negative association with shot success, while defender distance and shot clock have positive associations. The clutch coefficient is negative under both priors, suggesting that clutch situations are related to lower shot success rate with other factors controlled.

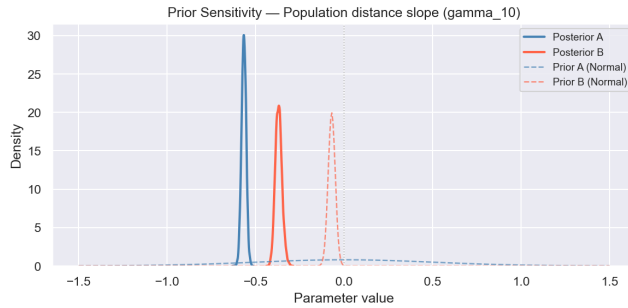


Figure 4: Prior sensitivity comparison for μ_{β_1}

However, the magnitude of some coefficients still changes under Prior B. For

instance, the shot distance effect shows the largest difference in magnitude between the two priors, as shown in Figure 4. Under Prior A, the posterior mean is -0.5665, while under Prior B it is -0.3697. This indicates that the stronger prior specification pulls the distance effect closer to its prior expectation. However, the effect remains clearly negative under both priors, so the substantive conclusion that longer shots are less likely to be made is unchanged.

Defender distance also remains positive under both priors, despite that the estimated magnitude is smaller under Prior B. Since other variables all maintain the same direction, the main regression conclusions are not driven by prior choice.

Overall, the sensitivity analysis suggests that the main conclusions are robust to choosing different priors. Prior B affects magnitude of posterior estimates without changing the directions or substantive model interpretation.

4.6 Model comparison

This section evaluates which prior specification generated better predictive performance. The two models have same likelihood and hierarchical structure, but differ in their prior distributions, which Prior B more informative.

Models are compared using leave-one-out cross validation (LOOCV). LOOCV estimates the expected log pointwise predictive density, denoted as $ELPD_{LOO}$. Higher $ELPD_{LOO}$ value indicates better expected predictive performance on new observations.

Model	$ELPD_{LOO}$	p_LOO	ELPD difference	Weight
Prior A	-64331.37	260.71	0.00	1.00
Prior B	-64398.23	286.52	66.86	0.00

Table 8: Model predictive power comparison

From Table 8, it can be noted that Prior A has better LOOCV performance by a higher $ELPD_{LOO}$ value and lower ELPD difference value. The model weight is also assigned entirely to Prior A.

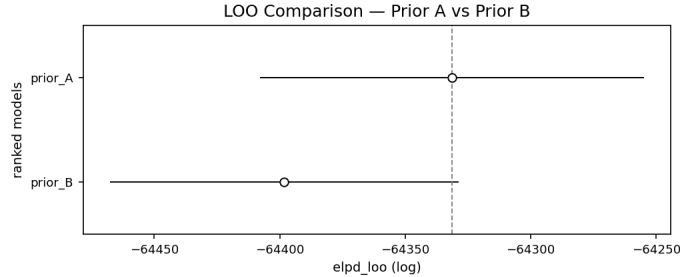


Figure 5: LOOCV comparison between Prior A and Prior B

In short, model comparison prefers the use of Prior A to Prior B. This suggests that weakly informative prior may allow the data to better determine the posterior estimates and generate more precise out-of-sample predictions.

4.7 Predictive validation

Predictive validation was conducted using the held-out test set. The cleaned dataset was split into training and testing sets using an 80/20 split. The fitted posterior samples were then used to predict the probability of shot success for each observation in the test set.

For each test shot, the model computed a posterior-average predicted probability $\hat{p}_i$. This was obtained by calculating the predicted make probability for each posterior draw and then averaging across posterior samples. The predicted probabilities were compared with the observed shot outcomes in the test set using different evaluation metrics.

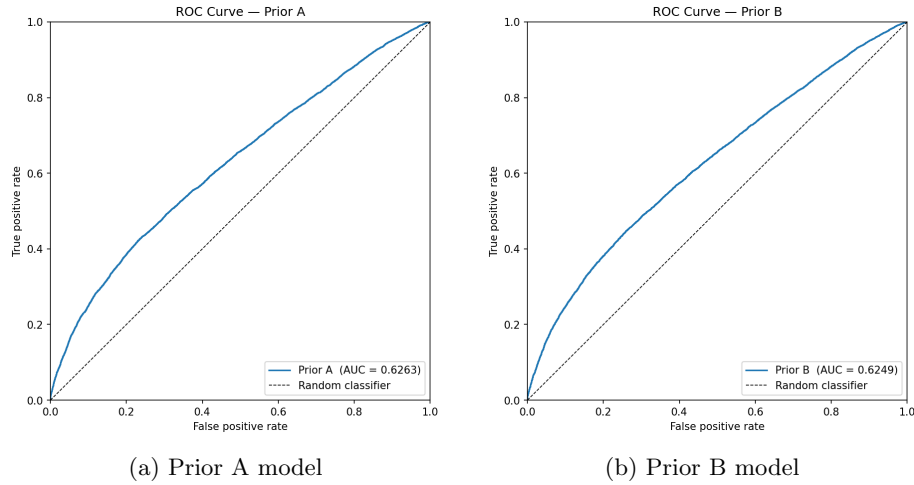


Figure 6: ROC curves of Prior A and Prior B models

Model	AUC	Brier Score	Log Loss
Prior A	0.6263	0.2354	0.6632
Prior B	0.6249	0.2356	0.6637

Table 9: Out of sample evaluations

Prior A outperform Prior B with slightly higher AUC, lower Brier score and log loss, which is consistent with LOOCV model comparison in Section 4.6.

Moderate predictive power is suggested by an AUC of 0.6263, which is reasonable since individual shot outcome contain substantial randomness and are

affected by factors not included in the model. Besides, individual coefficient and intercept may change overtime at different time points, even within the same season.

4.8 Summary of findings

Overall, findings in the Section 4 show that the Bayesian multilevel logistic regression model converged well under primary prior and produced interpretable posterior estimates. The convergence diagnostics suggest that MCMC chains mixed well and posterior samples were reliable for interpretation.

Posterior regression result examined hypotheses mentioned by this study and support main expectations. It directly addressed the five research hypotheses: posterior estimates under Prior A support all five hypotheses:

Hypothesis	Result	Conclusion
H1: Shot distance is negatively associated with FG%	$\mu_{\beta_1} = -0.5665$; 94% HDI below 0	Supported
H2: Closest defender distance is positively associated with FG%	$\beta_2 = 0.3140$; 94% HDI above 0	Supported
H3: Clutch-time shots have lower FG%	$\beta_4 = -0.1430$; 94% HDI below 0	Supported
H4: Player-level random intercept variance is greater than 0	σ_{β_0} SD = 0.1012; 94% HDI above 0	Supported
H5: The effect of distance varies across players	σ_{β_1} SD = 0.1382; 94% HDI above 0	Supported

Table 10: Hypothesis, result and conclusion

These findings confirm that shot context and player heterogeneity both matter when explaining NBA shot success. Longer shot distance decreases the probability of making a shot, while wider defender separation increases it. Clutch status is associated with lower shot success when other predictors are controlled. The model also detected meaningful player-level variation in both baseline shooting capacity and sensitivity to distance, supporting the multilevel structure.

Prior sensitivity analysis hints that these main conclusions are robust to different priors. Although Prior B influences estimates' magnitude, directions of key effects remain consistent across both priors. Model comparison using LOOCV and predictive validation both prefer Prior A to Prior B, letting Prior A serve as the primary model.

Taken together, the analysis results show that the model provides a useful balance between interpretability and predictive power. It captures individual level differences, identifies shot-quality factors to give Bayesian uncertainty estimates for NBA shot success.

5 Conclusion

This project applied a Bayesian multilevel logistic regression model on 121,617 shot attempts from 274 players in the 2014-15 NBA regular season. This model successfully identified the crucial factors influencing shot success and quantified the heterogeneous abilities across players, with all five research hypotheses supported under both Prior A and Prior B.

An interesting finding generated in this research is that under both priors, Steven Curry is the player with the highest baseline shooting ability among all players, which is consistent with the public belief. Notably, a substantial gap separates Curry from the second-ranked player, suggesting that his shooting advantage is not only real, but unusually large relative to the rest of the league.

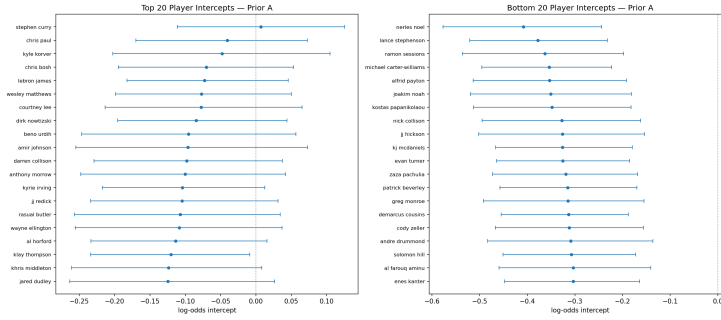


Figure 7: Trace Plot for Prior B

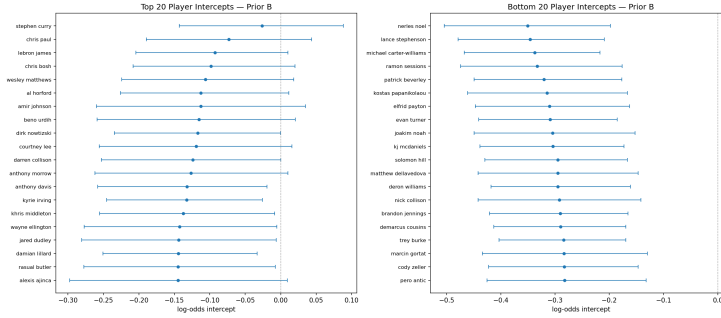


Figure 8: Trace Plot for Prior B

Overall, the Bayesian multilevel logistic regression framework provides a meaningful balance between interpretability and predictive power, integrating domain knowledge with data and offering posterior distributions for every quantity of interest. This model serves as a solid foundation upon which further studies on spatial, temporal, and contextual effects can be extended.

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